

ANALYSIS METADATA & COORDINATE REFERENCE SYSTEM

Study Area: Saswad Watershed, Maharashtra | CRS: EPSG:32643 — WGS 84 / UTM zone 43N
Spatial Resolution: 10 m × 10 m | Extent: 37.21 km² (3,72,099 total analyzed pixels)

1. SATELLITE METHODOLOGY & SOFTWARE ARCHITECTURE

- Satellite Sensor: European Space Agency (ESA) Sentinel-2 Multi-Spectral Instrument (MSI) Level-2A surface reflectance rasters.
- Spectral Indices Calculated:
 - NDVI: (B08 - B04) / (B08 + B04) utilizing Band 8 (NIR, 842 nm) and Band 4 (Red, 665 nm) at 10m ground resolution.
 - NDWI: (B03 - B08) / (B03 + B08) utilizing Band 3 (Green, 560 nm) and Band 8 (NIR, 842 nm) to delineate open water bodies.
- Spatial GIS Processing: Analysis performed via standardized QGIS raster calculation, thresholding, and zonal pixel statistics.
- Web GIS Architecture: The dashboard is built with React & TypeScript. The web application directly renders the GeoTIFF raster layers in the browser via geotiff.js onto Leaflet canvas overlays (GeoTIFF rasters are NOT converted to GeoJSON for map display). Verified summary metrics are strictly loaded from src/data/realMetrics.ts.

2. VERIFIED MEASURED ANALYSIS RESULTS (2023 vs 2026)

Target Indicator	2023 Baseline	2026 Current	Net Measured Change
Surface Water Area	12.53 ha (1,253 px)	3.03 ha (303 px)	-9.50 ha (-75.82%)
Positive-NDVI Vegetation Area	3,705.66 ha (370,566 px)	3,651.68 ha (365,168 px)	-53.98 ha (-1.46%)
Water Loss (2023 Water -> 2026 Non-Water)—		954 pixels detected	9.54 ha (95,400 m²)
Water Gain (2023 Non-Water -> 2026 Water)—		4 pixels detected	0.04 ha (400 m²)

LULC Class (2026 Analysis Extent)	Area (Hectares)	Area (Square Kilometers)	Share (%)
Vegetation (Positive NDVI)	3,651.65 ha	36.52 km²	98.14%
Other (Scrub, Barren, Fallow, Built-up)	66.31 ha	0.66 km²	1.78%
Surface Water (NDWI Positive)	3.03 ha	0.03 km²	0.08%
Total Analyzed Common Extent	3,721.00 ha	37.21 km²	100.00%

3. VERIFIED JALDRISHTI SYNTHESIS INSIGHTS

Insight 1: Detected Water-Area Contraction (Surface Water Extent)

Satellite analysis detected a significant contraction in surface water extent from 12.53 ha in 2023 to 3.03 ha in 2026 across the 37.21 km² analysis extent—a net reduction of 9.50 ha (-75.82%), driven by 9.54 ha of detected water loss against only 0.04 ha of localized water gain.

Insight 2: Vegetation Stability & Positive NDVI Coverage

Positive-NDVI vegetation cover exhibited high overall spatial stability with a slight reduction from 3,705.66 ha in 2023 to 3,651.68 ha in 2026, reflecting a marginal net decline of 53.98 ha (-1.46%) while remaining the dominant land-cover class across the watershed extent.

Insight 3: Watershed Decision-Support & Conservation Value

The dashboard provides evidence-based decision support by spatially localizing critical water depletion hotspots and stable vegetation canopies, enabling conservation authorities to prioritize field interventions, check dam desiltation, and artificial recharge structures where water loss is most acute.

4. PRESENTATION & VIVA DISCLOSURES (ACADEMIC FRAMEWORK)

A. Verified Measured Analysis Results

- All surface water and vegetation statistics represent empirical measurements computed via QGIS from Sentinel-2 rasters at 10m ground resolution.
- No simulated or synthetic multipliers have been added to the measured 12.53 ha (2023) and 3.03 ha (2026) water boundaries or the 3,705.66 ha (2023) and 3,651.68 ha (2026) vegetation covers.

B. Prototype Limitations

- The multi-temporal comparison observes two specific satellite acquisition timestamps (2023 baseline vs 2026 current). Inter-annual seasonal fluctuations require continuous dense time-series compositing.
- Field observations and water body markers currently displayed on the map serve as curated validation points rather than an exhaustive ground-truthing survey of every hydrological structure.

C. Future & Proposed Features (Roadmap)

- Automated imagery ingestion pipeline connecting to Copernicus Open Access Hub for automated cloud-free mosaic generation.
- Machine-learning predictive modeling for seasonal surface runoff estimation and sediment transport risk zones.
- Integration of physical IoT water-level telemetry sensors in check dams and percolation tanks for real-time validation.